Assignment 1 (M)

(Continuity, Differentiability, Differentiation)

Q1. A function $f(x)$ is defined by

$$f(x) = \begin{cases} \frac{[x^2]-1}{x^2-1}, & x^2 \neq 1 \\ 0, & x^2 = 1 \end{cases}$$ where $[\cdot]$ denotes GIF

1) is continuous at $x=-1$ 2) is discontinuous at $x=1$
3) is differentiable at $x=1$ 4) NOT.

Q2. If $f(x) = \text{Sgn}(x)$ and $g(x) = x(1-x^2)$ then $(fog)(x)$ is discontinuous at
1) exactly one point 2) exactly two points
3) exactly three points 4) no point.

Q3. If $f(x)$ is a polynomial satisfying the relation $f(x) + f(2x) = 5x^2 - 18$ then $f'(1)$ is equal to
1) 1 2) 3 c) cannot be found since degree of $f(x)$ is not given 4) 2

Q4. Let $f: R \to R$ be a differentiable function satisfying $f(y)\, f(x-y) = f(x)$ $\forall\, x, y \in R$ and $f'(0) = p$, $f(5) = q$ then $f(5)$ is
1) p^2/q 2) p/q 3) q/p 4) q.

Q5. The function $f(x) = \max\{3-x, 3+x, 6\}$ is differentiable at
1) All points 2) No point 3) All points except two points 4) All points expect at one point

Q6. The number of points of discontinuous of $g(x) = f(f(x))$ where $f(x)$ is defined as $f(x) = \begin{cases} 1+x, & 0 \le x \le 2 \\ 3-x, & 2 < x \le 3 \end{cases}$
1) 0 2) 1 3) 2 4) > 2

Q7. The function $f(x)$ is defined as $f(x) = \begin{cases} \frac{1}{|x|}, & |x| > 2 \\ a+bx^2, & |x| \le 2 \end{cases}$ where a & b are constants. then which of the

following is true ?

1) f is differentiable at $x=-2$ if and only if $a=3/4$, $b=-1/16$ 2) f is differentiable at $x=-2$ whatever be the values of a & b
3) f is differentiable at $x=2$ if $b=-\frac{1}{16}$ whatever be the values of a
4) f is differentiable at $x=-2$ if $b=\frac{1}{16}$, whatever be the values of a.

Q8. If $f(x)=\frac{[x]}{|x|}$, $x\neq 0$ where $[\cdot]$ denotes the GIF then $f'(1)$ is

1) -1 2) 2 3) ∞ (4) does not exist.

Q9. Define $f:[0,\pi]\to R$ by

$$f(x)=\begin{cases} \tan^2 x\left[\sqrt{2\sin^2 x+3\sin x+4}-\sqrt{\sin^2 x+6\sin x+2}\right], & x\neq\frac{\pi}{2} \\ k, & x=\frac{\pi}{2}\end{cases}$$

is continuous at $x=\frac{\pi}{2}$ then $k=$

1) $\frac{1}{2}$ 2) $\frac{1}{6}$ 3) $\frac{1}{24}$ 4) $\frac{1}{32}$.

Q10. Let $|a_1\sin x+a_2\sin 2x+\cdots+a_8\sin 8x|\le|\sin x|$ for $x\in R$

Define $p=a_1+2a_2+3a_3+\cdots+8a_8$. then p satisfies

1) $|p|\le 1$ 2) $|p|<1$ 3) $|p|>0$ 4) $|p|>1$

Q11. If $f(x)=\sec^{-1}([1+\sin^2 x])$; where $[\cdot]$ GIF. then the set of all points where $f(x)$ is not continuous is

1) $\{\frac{n\pi}{2}, n\in I\}$ 2) $\{(2n-1)\frac{\pi}{2}, n\in I\}$
3) $\{(n-1)\frac{\pi}{2}, n\in I\}$ 4) $\{n\pi / n\in I\}$

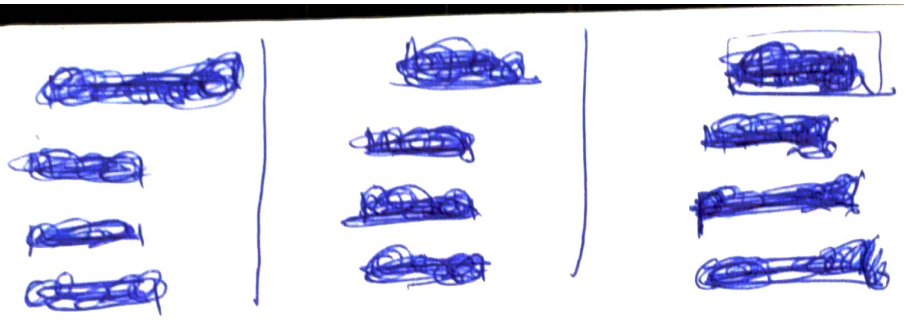
Q12. The function

$f(x)=(x^2-1)|x^2-3x+2|+\cos(|x|)$ is not differentiable at $x=$

1) -1 2) 0 3) 1 4) 2

Q13. The number of points in $(1,3)$ where $f(x)=a^{[x^2]}$, $a>1$ is not differentiable where $[\cdot]$ denotes the ~~GIF~~ integral part of x is

1) 0 2) 3 3) 5 4) 7



Q14. If $[\cdot]$ denotes the integral part of x and $f(x) = [x]\left\{\frac{\sin\frac{\pi}{[x+1]} + \sin \pi[x+1]}{1+[x]}\right\}$, then

1) $f(x)$ is continuous everywhere in its domain
2) $f(x)$ is continuous but not differentiable
3) $f'(x)$ exists for all x in its domain
4) $f(x)$ is discontinuous at all integral points in its domain.

Q15. Let $f(x) = [\cos x + \sin x]$, $0 < x < 2\pi$, where $[\cdot]$ denotes the GIF, then the number of points discontinuity of $f(x)$ is

1) 3 2) 4 3) 5 4) 6.

Q16. If the function $f(x) = -4e^{\frac{1-x}{2}} + 1 + x + \frac{x^2}{2} + \frac{x^3}{3}$ be an invertible function and $g(x) = f^{-1}(x)$, then the value of $g'\left(-\frac{7}{6}\right)$ equals

1) $\frac{1}{5}$ 2) $-\frac{1}{5}$ 3) $\frac{6}{7}$ 4) $-\frac{6}{7}$

Q17. Let a function $f(x)$ be such that $f''(x) = f'(x) + e^x$ & $f(0) = 0$, $f'(0) = 1$, then $\ln\left(\frac{f(2)^2}{4}\right)$ equals to

1) $\frac{1}{2}$ 2) 1 3) 2 4) 4

Q18. Let $f: (0, \infty) \to R$ and $F(x) = \int_1^x f(t)\,dt$. If $F(x^2) = x^2(1+x)$ then $f(4)$ equals

1) 5/4 2) 7 3) 4 4) 2

Q19. If $f(x) = \int_0^x (1+t^3)^{-\frac{1}{2}}\,dt$ be an invertible function & $g(x)$ is the inverse of f, then the value of $\frac{g''(x)}{g(x)}$ is

1) 3/2 2) 2/3 3) 1/3 4) 1/2.

Q 20. If $f(x)$ be positive, continuous and differentiable on the interval (a,b). If $\lim_{x\to a^+} f(x) = 1$ and $\lim_{x\to b^-} f(x) = 3^{1/4}$ also $f'(x) > (f(x))^3 + \frac{1}{f(x)}$, then

1) $b-a > \frac{\pi}{24}$ 2) $b-a < \frac{\pi}{24}$ 3) $b-a = \frac{\pi}{12}$ 4) $b-a = \frac{\pi}{24}$.

Q 21. If a function $f: [-2a, 2a] \to R$ is an odd function such that $f(x) = f(2a-x)$ for $x \in [a, 2a]$ and the left hand derivative at $x = a$ is zero then left hand derivative at $x = -a$ is ____

1) a 2) 0 3) $-a$ 4) 1

Q 22. $f: R \to R$ is one-one, onto and differentiable function and graph of $y = f(x)$ is symmetrical about the point $(4,0)$, then which of the following is correct ?

1) $f^{-1}(2010) + f^{-1}(-2010) = 8$ 2) $\int_{-2010}^{2018} f(x)dx = 2018$

3) If $f'(-100) > 0$, then roots of $x^2 - f'(10)x - f'(10) = 0$ may be non-real

4) If $f'(10) = 20$ then $f'(-2) = 12$.

Q 23. Let f be a function defined by $f(x) = 2^{|\log_2 x|}$, then at $x = 1$

1) $f(x)$ is continuous as well as differentiable

2) Continuous but not differentiable

3) differentiable but not continuous

Q 24. Let $f(x+y) = f(x)\,f(y)$ for all x & y. Suppose that $f(3) = a$ and $f'(0) = b$ then $f'(3)$ is given by $(a, b \in R)$

1) ab 2) $a+b$ 3) $2ab$ 4) $2a+b$

Q25. Number of points where the function $f(x) = \max(|\tan x|, \cos|x|)$ is non-differentiable in the interval $(-\pi, \pi]$ is

1) 4 2) 6 3) 3 4) 2

Q26. If $f(x) = [x](\sin kx)^p$ is continuous for real x, then

1) $k \in \{n\pi, n \in I\}, p > 0$ 2) $k \in \{\frac{n\pi}{2}, n \in I\}, p > 0$

3) $k \in \{n\pi, n \in I\}, p \in R - \{0\}$

4) $k \in \{n\pi, n \in I, n \neq 0\}, p \in R - \{0\}$

Q27. $f(x) = \begin{cases} x+2, & x < 0 \\ -x^2 - 2, & 0 \le x < 1 \\ x, & x \ge 1 \end{cases}$. then the number of points of discontinuity of $|f(x)|$ is

1) 1 2) 2 3) 3 4) NOT.

Q28. $f(x) = \begin{cases} 3 - \left[\cot^{-1}\frac{2x^3 - 3}{x^2}\right], & \text{if } x > 0 \\ \{x^2\}\cos(e^{1/x}), & \text{if } x < 0 \end{cases}$ is continuous at $x = 0$. then the value of $f(0)$, where $[\cdot]$ & $\{\cdot\}$ denote the GIF & fractional part functions respectively is

1) 0 2) 1 3) -1 4) NOT.

Q29. If $f(x) = \begin{cases} x^2(e^{1/x}), & x \neq 0 \\ k, & x = 0 \end{cases}$ is continuous at $x = 0$ then

($\{\cdot\}$ denotes fractional part function)

1) It is differentiable at $x = 0$

2) $k = 1$

3) continuous but not differentiable at $x = 0$

4) continuous everywhere in its domain.

Q30. Let $f(x) = \cos(x \cos\frac{1}{x})$

ST: $f(x)$ is discontinuous at $x = 0$

ST: $\lim_{x \to 0^+} f(x)$ does not exist

1) 1 2) 2 3) 3 4) 4.